

AI-based Tooling – Near Future



Work in Progress

- ISO WG-14 responsible for standards such as
 - ISO/PAS 8800:2024 : Road vehicles — Safety and artificial intelligence
 - ISO 22440 (draft) : Artificial intelligence — Functional safety and AI systems
In particular: Appendix C about AI Tool Classification and Qualification
- EUROCAE WG-114 / SAE G-34
 - ER-022 Statement of concerns published in 2021
<https://www.eurocae.net/product/er-022-artificial-intelligence-in-aeronautical-safety-related-systems-statement-of-concerns/>
 - ED-324 / ARP6983 (publication planned for the end of 2026)

From EASA Concept Paper Issue 3

- <https://www.easa.europa.eu/en/document-library/general-publications/easa-artificial-intelligence-concept-paper-issue-2>
- Issue 3 for comments: <https://www.easa.europa.eu/en/downloads/143701/en>

AI technology	Limit in assurance level allocation to AI constituents		
	IDAL (initial airworthiness)	AL/SWAL (ATM/ANS)	AL / TQL (AI & tool qualification standards)
Supervised learning	IDAL C	AL 3 / SWAL 2	AL 3 / TQL 3
Unsupervised learning	IDAL D	AL 5 / SWAL 4	AL 5 / TQL 5
Reinforcement learning	IDAL C	AL 3 / SWAL 2	AL 3 / TQL 3
Logic- and knowledge-based	IDAL C	AL 3 / SWAL 2	AL 3 / TQL 3
Large OTS models	IDAL D	AL 5 / SWAL 4	AL 5 / TQL 5

Table 2 — Assurance level limitations by AI technology

Comments ?
Ideas ?



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